ML Task 1 Report

House Price Prediction using Linear Regression

1.Introduction

Machine Learning is a technique that enables computers to learn from data and make predictions. In this project, Linear Regression is used to predict house prices based on different features. Linear Regression is a supervised learning algorithm that finds a relationship between input variables and a continuous output variable.

2.Dataset Description

The California Housing dataset is used in this project. It contains information about houses such as:

- Median Income
- House Age
- Number of Rooms
- Population
- Location details

The target variable is the **house price**.

3.Exploratory Data Analysis (EDA)

The dataset was explored using:

- head() to view data
- info() to check data types
- describe() for statistics

Observations:

- No missing values found
- Median income has a strong impact on house price

- Data is numerical and suitable for regression

Graphs like scatter plots were used to visualize relationships.

4. Data Preprocessing

- Checked for missing values (none found)
- Selected features (input variables)
- Split dataset into:
    - Training set (80%)
    - Testing set (20%)

5.Model Used

Linear Regression model from scikit-learn was used.

Steps:

- Model initialized
- Trained using training data
- Predictions made on test data

6. Results and Evaluation

The model was evaluated using:

- **MAE (Mean Absolute Error)**: measures average error
- **RMSE (Root Mean Square Error)**: penalizes large errors
- **R^2 Score**: shows accuracy of model

Example results:

- MAE: 0.53
- RMSE: 0.72
- R^2 Score: 0.60

7. Conclusion

The Linear Regression model successfully predicted house prices with reasonable accuracy. The workflow of machine learning was followed properly, including data analysis, model training, and evaluation. The results show that the model performs well but can be improved further.

8. Future Improvements

- Use advanced models (Decision Tree, Random Forest)
- Perform feature engineering
- Hyperparameter tuning
- Add more real-world data